A leakage-controlled test for incremental predictive information in neurophysiological signals

Current state, validated apparatus, and complete findings inventory

Kahlus / Neural-CASP

Prepared 25 July 2026 for Todd P. Coleman and Amrith Lolitkar

Repository `Kahlus-V1`, branch `fix/forecast-overlap-guard`

Statement of scope. This document reports what the Kahlus evaluation apparatus has been demonstrated to do, with every statistic traceable to a named artifact in the repository. **It reports no positive biological finding, because none exists.** Every real-data result obtained to date is a negative or an underpowered null, and each is labelled as such below. The results we consider substantive are methodological: a calibrated detection floor (Section 3), a quantified evaluation-leakage failure mode (Section 4), and one instance in which the control battery rejected a result that the bootstrap confidence interval alone would have certified (Section 5). We regard the last of these as the most useful single result the project has produced.

1 The estimand

For an outcome Y at horizon h, a nuisance block B, and a candidate block Z, we define the *residual forecastability score*

$$\mathrm{RFS}_{\mathrm{bits}} = \frac{\mathrm{NLL}(Y \mid B) - \mathrm{NLL}(Y \mid B, Z)}{\ln 2}, \tag{1}$$

where NLL is the mean out-of-fold binary cross-entropy of a calibrated probability forecast, with probabilities clipped to $[10^{-5}, 1-10^{-5}]$. Implementation: `forecastability/m1.py::_rfs_bits`.

Equation (1) is the out-of-sample log-likelihood gain from adding Z to B, in bits per anchor. Methodologically it is a cross-fitted, subject-held-out analogue of nested ΔR^2 / $\Delta \mathrm{CV}R^2$ model comparison — the same incremental-value question posed in Rao et al. (2025, §2.5.2), evaluated out-of-sample under a mandatory control battery rather than in-sample.

What RFS is not. RFS is *not* a mutual-information estimate, and we do not interpret it as one. McAllester and Stratos (2018) prove that no distribution-free high-confidence lower bound on mutual information from N samples can exceed $O(\log N)$; any estimator read as a certified MI bound inherits that limitation. RFS is a difference of out-of-fold mean log-losses between two nested models with bounded predictions — a bounded functional of held-out forecasts. Concentration and conformal arguments apply to *that* quantity. Reading RFS as a bound on $I(Y; Z \mid B)$ is not licensed by our procedure.

The blocks are fixed across all findings below:

- B (nuisance) always contains the event base rate, recent event history, and cortical spectral features. A positive RFS therefore means Z carries information the cortex's own recent state does not already supply.
- Z (candidate) is the block under test.
- Y is a future event indicator at horizon h.

2 Evaluation protocol

2.1 Cross-fitting and clustering

Nuisance and residual models are fit out-of-fold with `GroupKFold` at the *subject* level. All resampling is at the cluster (subject) level, never per-anchor, since anchors within a recording are strongly dependent.

2.2 Baseline ladder

The residual is scored against the *minimum-NLL* member of a candidate baseline set (moving average, persistence, base rate, cross-fitted nuisance-only model). Reporting a gain over a weak baseline is a standard route to inflated incremental-value claims; selecting the strongest baseline removes that degree of freedom.

2.3 Control battery

Every candidate result must survive all of:

Control	What it breaks
Label shuffle	Any association at all
Time shift	Temporal alignment
Within-subject circular shift of Z	Alignment, preserving Z's autocorrelation
Subject-identity probe	Z acting as a subject fingerprint
Cluster permutation	Supplies the p-value

2.4 Certification rule

Implemented in `_rfs_eval.py::horizon_real_passes`. An edge is certified only if *all* of the following hold:

$$\begin{aligned} \mathrm{RFS}_{\mathrm{bits}} &> 0.02 \\ \mathrm{LCB} &> 0 \quad \text{(one-sided 95\% subject-cluster bootstrap)} \\ \text{every control} &< 0.40 \times \mathrm{RFS}_{\mathrm{bits}} \\ \text{nuisance probe failures} &= \varnothing \end{aligned}$$

plus minimum subject and minimum positive-event counts. The 0.02 bits floor is a detection threshold, not a claim of clinical utility.

3 Validated apparatus: calibration

Detection power is a rate over seeds; a single run per condition cannot estimate it. We inject the candidate coupling at graded strengths into a synthetic fixture with known ground truth and run the complete gate at each strength across 24 independent seeds.

Table 1: Detection-power sweep, 24 seeds per point, synthetic ground truth. Detection = fraction of seeds whose one-sided 95% bootstrap lower bound clears zero. Source: `runs/autonomic_calibration/detection_power_sweep.csv`.

Injected coupling	Detection power	Detections	Median RFS (bits)
0.0 (null anchor)	**0.042**	1 / 24	−0.0005
0.5	0.583	14 / 24	+0.0084
1.0	**0.792**	19 / 24	+0.0211
2.0	0.667	16 / 24	+0.0145

False-positive rate. The null anchor fires on 1 of 24 seeds, = 4.2%, against a nominal 5% for a one-sided 95% lower bound. The single false positive (seed 2) has RFS = 0.0022 bits with LCB = 0.00023 — barely clearing zero, which is the expected appearance of a correctly calibrated borderline detection rather than a malfunction. Per-seed values are retained in `detection_power_per_draw.csv` so individual detections can be audited.

Non-monotonicity is real and is explained. Power declines from 0.792 at strength 1.0 to 0.667 at strength 2.0, and median bits from +0.0211 to +0.0145. This persists across all 24 seeds in both panels and is not seed noise. RFS is a *residual*: once the outcome becomes highly predictable from base rate and recent history alone, B absorbs the signal and the residual attributable to Z shrinks. Detection power is therefore the calibration quantity we report; median recovered bits is shown alongside it and is explicitly not a calibration curve.

Scope. These figures are synthetic by construction. They establish that the estimator is calibrated before it is pointed at real data. They are not evidence about any cohort.

4 Validated apparatus: a quantified leakage failure mode

The project's original headline result was an artifact, and the mechanism is worth reporting because it survives an otherwise correct subject-held-out split.

The historical forecasting task took a 128-sample window, used X = seq[0:127] as input and Y = seq[1:128] as target, and scored mean squared error over the *full* 127-length target. Input and target therefore shared **126 of 127 samples**. A model that copies its input shifted by one is exact on 126 positions; the single genuinely future sample is averaged away.

Distinction from the existing leakage literature. Brookshire et al. (2024) and Del Pup et al. (2025) document *train/test* dependence: related windows crossing folds. Reported inflation in that literature is of order 8–13% F1 (Dehghani et al., 2019) up to 28% accuracy

Table 2: Identical ridge model, identical real Sleep-EDF data, two scoring rules. Source: `docs/paper/ieee_figure_source/data/amrith_overlap_headline.json`.

Scoring rule	Ridge MSE
Full 127-sample target (126/127 shared with input)	0.00180
Isolated strictly-future sample	0.22694
Inflation factor	**126×**

(Kattenborn et al., 2022). The failure above is different in kind — it is *within-window* input/target overlap, which survives a perfectly correct subject-held-out split and is invisible to any split-level audit. That is why its magnitude is two orders larger.

Remediation, and how it was verified. Forecasting tasks now carry a boolean `metric_mask` true only on target positions absent from the input window, applied to both the training loss and the evaluation metric. The regression test constructs a predictor that exploits the overlap: it scores 0.059 unmasked and 0.470 masked, an **8× discriminating gap**. It is that gap, not agreement between two metrics, that demonstrates the mask functions.

A guard (`data/forecast_contract.py`) makes the unsafe configuration unconstructible: a forecasting task is valid only if horizon $\geq$ window length or a non-empty strictly-future mask is attached. The historical claim is marked invalid in place, not deleted.

Person-level split auditing. A separate audit addresses a distinct failure: declared-`subject_id` checks trust their labels, so when two nights of one person are labelled as two subjects the check passes while the person sits on both sides of the split. The regression test constructs exactly that manifest and asserts that the conventional check *passes* while the person-level check *fails*. Where person identity cannot be established the audit fails closed.

5 Result: the control battery rejects a bootstrap-certifiable effect

This is, in our assessment, the most substantive result the project has produced.

We ran the gate on MIT-BIH Polysomnographic Database (slpdb) — 11 records, 9 subjects, 6152 windows, claim-grade bootstrap, subject-held-out — with Z an autonomic block (QRS-derived heart rate, blood-pressure and respiration log-power envelopes), B cortical spectral features plus event history, and Y a future arousal indicator. Arousals were stratified into respiratory (hypopnea/apnea-terminating) and non-respiratory (bare and leg-movement) classes, because the precedence literature diverges sharply between them.

The respiratory rows would have been reported as positive. At $h = 2$ and $h = 4$ the subject-cluster bootstrap lower bound clears zero, and the cluster permutation test returns $p = 0.031$ at $h = 2$ (512 permutations, patient-cluster sign-flip, exact mode). On the confidence interval and p-value alone, these are findings.

They are rejected, on two independent grounds.

Table 3: slpdb autonomic gate, by arousal class and horizon. "Worst control" is the largest of the time-shift and circular-shift surrogates, expressed as a multiple of the real effect; the certification rule requires < 0.40.

Class	h	Pos. events	RFS (bits)	LCB	Worst ctrl / real	Subj. probe
Non-respiratory	1	190	−0.0265	−0.1186	—	fail
Non-respiratory	2	288	−0.0139	−0.0519	—	fail
Non-respiratory	4	408	−0.0236	−0.0502	—	fail
Respiratory	1	1360	+0.0058	−0.0022	1.42	fail
Respiratory	2	1795	+0.0087	**+0.0008**	**1.05**	fail
Respiratory	4	2170	+0.0111	**+0.0015**	**1.57**	fail

1. **The time-shift control scores at or above the real effect at every horizon** — 1.42×, 1.05×, 1.57×, against a required $< 0.40\times$. Destroying temporal alignment does not reduce the score. Whatever is being measured is therefore not predictive in time.
2. **The subject-identity probe fails at every horizon in both classes.** The autonomic block identifies which subject a window came from. With 9 subjects of differing apnea severity and correspondingly different arousal base rates, a subject fingerprint alone manufactures an apparent residual.

The effect is static, subject-level structure — not forecasting. **The bootstrap confidence interval would have certified it; the control battery rejected it.** We report this as direct real-data evidence that the controls are load-bearing rather than ceremonial, and as a concrete instance of what a CI-only pipeline would have shipped as a finding.

The non-respiratory arm is underpowered, not null. That class has 190 positive events at $h = 1$, and **73% of all non-respiratory arousals in the cohort (139 of 191) come from a single subject**; only 4 of 9 subjects have ≥ 10 such events. The $h = 1$ interval spans $[-0.119, +0.043]$ — consistent with anything from a large negative to a moderate positive. This is the class for which autonomic precedence is actually supported in the literature, and slpdb cannot test it. We report it as underpowered and draw no inference.

6 Complete findings inventory

Every real-data finding, with its status stated exactly.

Reading F2 correctly. Both learned models separate decisively from random and mean-of-trace at every horizon, and the margin over persistence widens with h — at $h = 64$ persistence has degraded to 2.084 while ridge holds 0.884. That is a real effect. But the GRU's advantage over a regularised linear autoregression is small and reverses at $h = 64$. At single-channel millisecond scale the signal is close to linear, and a ridge captures nearly all of it. This is consistent with the finding that well-regularised linear and Riemannian methods remain competitive on EEG (Lotte et al., 2018), and with recent foundation-model evaluations reporting no consistent advantage for larger architectures in low-data or transfer regimes.

Table 4: All findings. No entry is a positive biological result.

ID	Status	Result
F1	Invalidated artifact	Overlap-inflated forecasting MSE; 126×, marked invalid in place (Section 4).
F2	Negative (architecture)	Isolated strictly-future forecasting; learned models separate clearly from trivial baselines, but the GRU does not meaningfully beat a 127-tap ridge and loses at $h = 64$. See Table 5.
F3	Scout-grade negative	Peripheral channels → future stage transition, Sleep-EDF. Does not survive controls at scout thresholds. Design-limited: Sleep-EDF cassette carries no clean ECG.
F4	**Powered negative**	Complexity beyond spectral, 78 subjects, 413,828 windows, $n_{\text{boot}} = 2000$. Wake -0.330 $[-0.349, -0.313]$; NREM -0.284 $[-0.297, -0.271]$; REM -0.161 $[-0.177, -0.146]$. All CIs exclude zero on the negative side: the complexity block actively hurts.
F5	**Powered negative**	Propofol sedation, OpenNeuro ds005620, full 21/21 cohort, 2527 windows. RFS -0.2072 $[-0.3084, -0.1208]$. Same conclusion as F4 under a different etiology. Not TMS-evoked PCI; no consciousness or clinical claim.
F6	Built, unrun	NSRR MESA/SHHS. Synthetic known $+0.021$ / null -0.000 passes; real cohort blocked on credentialed access.
F7	Negative + underpowered	slpdb, Section 5. Respiratory: caught false positive. Non-respiratory: underpowered.

7 What the negatives do and do not establish

We state the limitations plainly, because several materially weaken the interpretation of the results above.

1. **No positive real-data finding exists.** F2–F5 are negative; F6 is unrun; F7 is negative in one class and underpowered in the other.
2. **The powered negatives are on targets where a null was the expected outcome.** Sleep-stage discrimination is well served by spectral features, so complexity adding nothing is unsurprising. This weakens the rhetorical force of F4 and F5 even though it does not weaken their validity.
3. **F4 and F5 test one specific complexity block**, not "complexity" in general.
4. **There is a published null against the broad autonomic hypothesis.** Rao et al. (2025) tested cardiac autonomic predictors in a nested design and found $\Delta R^2 < 0.03$, $p > 0.25$ (HRV RMSSD partial $r = 0.16$, $p = 0.29$) for subjective sleep quality. Our defence is timescale and outcome specificity — a within-night, event-locked arousal at a fixed horizon rather than a night-level subjective rating — and that is a defence, not a

Table 5: F2 — isolated single-future-sample forecasting MSE, subject-held-out Sleep-EDF, every model trained directly per horizon. Lower is better; best per row in bold. Source: `runs/amrith_isolated_check/sweep.csv`.

h (samples @100 Hz)	Random	Mean-of-trace	Persistence	Ridge-AR	GRU
1	1.855	1.037	0.281	0.227	**0.211**
2	1.865	1.081	0.338	0.310	**0.277**
4	1.981	1.155	0.488	0.430	**0.400**
8	2.070	1.232	0.853	0.675	**0.583**
16	2.054	1.264	1.334	0.822	**0.690**
32	2.254	1.432	1.855	0.956	**0.883**
64	2.208	1.395	2.084	**0.884**	0.913

refutation.

5. **slpdb is an apnea cohort**, so for the respiratory class it is the sub-population where autonomic precedence is *least* supported. It is an adversarial case, not a small version of MESA.
6. **Calibration evidence is synthetic.** It bounds the estimator's operating characteristic under a generative model we wrote. It is not external validation.
7. **RFS is not monotone in effect size** (Section 3), so detection power, not recovered bits, is the calibration quantity.

8 On the relationship to Rao et al. (2025)

We wish to be precise about this, since we cite the work as methodological grounding and do not wish to misrepresent it.

The positive result in Rao et al. is *intrinsic* infraslow rhythmicity — EGG ISO amplitude variance predicting subjective sleep quality ($\Delta R^2 = 0.13$, $p = 0.014$, $n = 49$ nights). The *coupling* measures added essentially nothing (κ: $\Delta R^2 = 0.0005$, $p = 0.84$; parietal MI: $\Delta R^2 = 0.046$, $p = 0.17$), and the paper's own conclusion attributes the effect to consistency of infraslow modulation within each organ rather than to stomach–brain coupling. Citing that paper as motivation for a *coupling* claim would misread it, and we do not.

What we take from it is the nested-model design of §2.5.2: a baseline model of macro-architectural and covariate features, then the candidate block added and partial R^2 reported. Our difference is cross-fitting and subject-held-out evaluation in place of in-sample fit, plus a mandatory control battery. We note that the paper already reports a cross-validated $\Delta\mathrm{CV}R^2$ (-0.015 for the phase-coupling measure), so the CV-shrinkage concern is acknowledged in that work.

9 Immediate requirements

1. **NSRR credentialed access (MESA, then SHHS).** slpdb has settled nothing about the F6/F7 question, for the concrete reason given in Section 5: the arousal class the hypothesis concerns has 190 events, 73% from one subject. MESA is required, not preferable. SHHS would then provide a dataset-held-out replication — an axis almost

nobody evaluates, and where the published cross-cohort drop is 0.12–0.32 absolute F1 (Ehrlich et al., 2024).

2. **Separation from arousal *detection*.** ECG-only arousal detection already reaches AUROC 0.93 on MESA (Li et al., 2020) and AUC 0.91 from five post-onset beats (Basner et al., 2007). Our estimand is strictly-future forecasting conditioned on a cortical baseline. Any presentation that does not make this explicit will be read as a weaker version of existing detection work.

10 Reproduction

```
python3.11 -m venv .venv && .venv/bin/pip install -e '.[figures,moabb]' pytest edfio

PYTHONPATH=src python -m pytest -q            # 893 passed, 4 skipped, 114 subtests
uvx ruff@0.16.0 check .                       # clean

# calibration (Section 3)
PYTHONPATH=src python3 scripts/autonomic_rfs_calibration_sweep.py \
    --out runs/autonomic_calibration --seeds 24 --strengths 0 0.5 1 2

# slpdb gate (Section 5)
PYTHONPATH=src python3 scripts/run_slpdb_autonomic_gate.py \
    --out artifacts/slpdb_autonomic_non_respiratory --bootstrap-mode claim \
    --arousal-class non_respiratory --data-dir <local slpdb>
```

Every gate emits `claim_scope` and `stop_reason` into its JSON and Markdown artifacts, so a run's permitted interpretation travels with its numbers.

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